

Kavindu Damsith

Computer Science & Engineering | Full-Stack Development | Applied AI

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PROFILE

Computer Science & Engineering background with practical experience delivering full-stack web and mobile products, backend services, cloud-ready applications, and applied AI workflows. Work spans requirements analysis, responsive interface development, REST API design, authentication and role-based access, payment integration, relational and document databases, real-time features, testing, and containerised deployment. Commercial, freelance, internship, and academic projects include a live Canadian business platform, a UK education product, multi-role mobile applications, healthcare systems, and real-time video research. Multiple competitive-programming and hackathon placements demonstrate disciplined problem-solving, teamwork, and the ability to deliver under time constraints.

EDUCATION

B.Sc. (Hons) in Computer Science & Engineering
University of Moratuwa, Sri Lanka

Requirements completed: Jun 2026

G.C.E. Advanced Level — Physical Science
3 A passes

2020

G.C.E. Ordinary Level
9 A passes

2017

TECHNICAL SKILLS

LANGUAGES

JavaScript, TypeScript, Python,
Java, PHP, C, C++, C#

FRONTEND & MOBILE

React, Next.js, React Native,
Flutter, Tailwind CSS, Material UI

BACKEND & DATA

Node.js, Express, Spring Boot,
Django, Flask, MySQL,
MongoDB, Sequelize

AI & ENGINEERING

PyTorch, TensorFlow, OpenCV,
YOLO, Git, Docker, Kubernetes,
Jenkins

PROFESSIONAL EXPERIENCE

Software Engineering Intern

Dec 2024 – Jun 2025

Cloud Solutions International | Sri Lanka

- Supported cloud application delivery and automated development workflows through Jenkins-based CI/CD pipelines.
- Built a multi-node Kubernetes environment for Docker-based services; configured HPA/VPA scaling, Prometheus monitoring, JVM tuning, and workload resource limits.
- Contributed testing and reliability improvements to Antikythera, an open-source Spring test-generation tool, by validating dependency-resolution behaviour and documenting edge cases.

Freelance Developer

2020–2024

Fiverr | Remote

- Delivered client projects across machine learning, data extraction and automation, and Arduino-based prototyping.
- Clarified requirements, developed and tested complete workflows, and managed client communication, revisions, and technical handover.

Web Developer

2020–2024

FutureX

- Built and maintained frontend and backend features for real-world web applications.
- Worked through Git-based collaboration, debugging, interface refinement, and application maintenance.

PROJECTS

Final-year research

ForkNetRT — Real-Time Video Super-Resolution

Research system for reconstructing sharper video while balancing latency, temporal consistency, and computational efficiency.

- Developed and evaluated a ForkNet-based PyTorch pipeline for resource-constrained, real-time operation.
- Built FFprobe metadata handling, native-FPS preview, CUDA/NVENC streaming, and runtime telemetry interfaces.

Python | PyTorch | OpenCV | FFmpeg | Flask | React

[Research profile](#) | [Engineering prototype](#)

Commercial | UK client | Private repository

Abacus Learning Platform

Education platform designed to help children build mental-calculation skills through structured, progressive learning.

- Implemented progressive lesson unlocking, dashboards, file delivery, payments, notifications, and real-time updates.
- Integrated Stripe payments, Socket.IO communication, and AWS S3 media storage.

React | Node.js | Express | MySQL | Socket.IO | Stripe | AWS S3

Open source | Internship contribution

Antikythera Test Generator

Java tooling that analyses Spring controllers and dependencies to support automated application-test generation.

- Tested dependency resolution across repositories, interfaces, DTOs, annotations, and dependency graphs.
- Documented edge cases and strengthened JavaParser-based unit and integration test coverage for Spring application structures.

Java | Spring | JavaParser | Maven | JUnit

[GitHub repository](#)

Competition project | Full-stack and AI

HealthLink

Healthcare platform combining patient and clinical workflows with brain-tumour MRI analysis.

- Developed patient, doctor, appointment, report, and administration workflows.
- Connected React and Node.js services to a Flask-based ResUNet pipeline and prepared Kubernetes deployment assets.

React | Express | MySQL | Flask | ResUNet | Kubernetes

[GitHub repository](#)

Commercial | Active development

Maple Wraps

Full-stack platform for a Canadian event-floor-wrap business, connecting product discovery, configurable quoting, customer accounts, projects, and operations.

- Built the React and TypeScript customer application, Next.js administration workspace, Express API, and MongoDB data layer.
- Implemented role-based access, configurable estimators, customer and team workflows, project tracking, guest-list processing, and QR table lookup.

React | TypeScript | Next.js | Express | MongoDB

[maplewraps.ca](#)

Mobile platform | Team project

PlateShare

Multi-role food-sharing platform connecting buyers, sellers or donors, delivery partners, and organisations coordinating donations.

- Primarily developed the buyer module and fully implemented organisation and delivery workflows.
- Integrated discovery, donations, live order state, maps, media, email, and Stripe payments.

React Native | Expo | Express | Sequelize | MySQL | Stripe

[GitHub repository](#)

Academic | Transactional system

Cinema Booking Platform

End-to-end theatre experience spanning movie discovery, scheduling, seat selection, secure checkout, refunds, and engagement features.

- Implemented temporary reservations, authentication and authorisation, reviews, comments, discounts, and promo codes.
- Added Stripe checkout, refund flows, Socket.IO updates, and chatbot assistance.

React | Express | Sequelize | MySQL | Stripe | Socket.IO

Commercial | Private repository

HyperTalk

Assistive communication product translating sign language to speech and speech back into an accessible visual workflow.

- Combined a Flutter client with Django services and a computer-vision pipeline.
- Integrated MediaPipe, OpenCV, TensorFlow, and text-to-speech; recognised as second runner-up at BrainStorm 2023.

Flutter | Django | TensorFlow | MediaPipe | OpenCV | TTS

ADDITIONAL PROJECTS

TrainGPS Safety Intelligence

Railway safety prototype combining ESP32 GPS capture, geospatial risk estimation from historical data, and real-time driver-attention monitoring.

Python | Jupyter | PHP | ESP32 | Geospatial ML
[Repository](#)

RouteOptima

Supply-chain application covering customer and shop operations, ordering, route assignment, payment records, and management dashboards.

React | Express | MySQL | Material UI
[Repository](#)

Feature Flag Platform Lab

Kubernetes comparison environment for FF4J, flagd/OpenFeature, Flagsmith, and Unleash using repeatable Kustomize conventions.

Kubernetes | Kustomize | OpenFeature | PostgreSQL

RPAL Interpreter

Language processor implementing lexical analysis, recursive parsing, AST generation and standardisation, environments, and CSE-machine evaluation.

Python | Parsing | AST | CSE Machine
[Repository](#)

OpenFeature Load Lab

Spring Boot experiment evaluating TLS-connected flagd feature flags under concurrent access through controlled batches of 4,000 lookups.

Java 17 | Spring Boot | OpenFeature | flagd | Maven
[Repository](#)

Kubernetes Scaling Challenge

Deployment work covering healthcare API workloads, HPA/VPA scaling, configuration separation, storage integration, and scripted rollouts.

Kubernetes | HPA | VPA | Kustomize | Shell

ENGINEERING HIGHLIGHTS

Product workflows

Authentication, role-based access, configurable quoting, payments, refunds, administration, and real-time user flows.

Applied AI integration

Computer-vision and deep-learning services connected to usable web, mobile, healthcare, and video workflows.

Backend and data

REST services, relational modelling, document data, validation, Sequelize, MySQL, MongoDB, and PostgreSQL.

Delivery and quality

Git-based collaboration, automated tests, Docker, Kubernetes, CI/CD, monitoring, and performance-aware implementation.

AWARDS & COMPETITIONS

1 **MoraXtreme 9.0**
First place, 2024, among more than 350 teams island-wide.

2 **Hackdoze 1.0**
First runner-up, 2021.

4 **ENIGMA 24 - Crack the Code**
Fourth place in the team finals, 2024.

1 **CYPHER 2023 Hackathon**
Winner, team competition, 2023.

3 **BrainStorm Innovation Competition**
Second runner-up, 2023, as part of the team behind HyperTalk.

8 **MoraXtreme 8.0**
Eighth place, 2023, among more than 300 teams.

REFEREES

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